Lab Report: Image classification experiments

1 Introduction

In this report we compare several models on standard benchmarks. All experiments were run on a single GPU with 24 GB of memory. The learning rate was set to 0.01 with the Adam optimizer.

2 Results

Figure 1 shows the f1 score of DistilBERT by dataset.

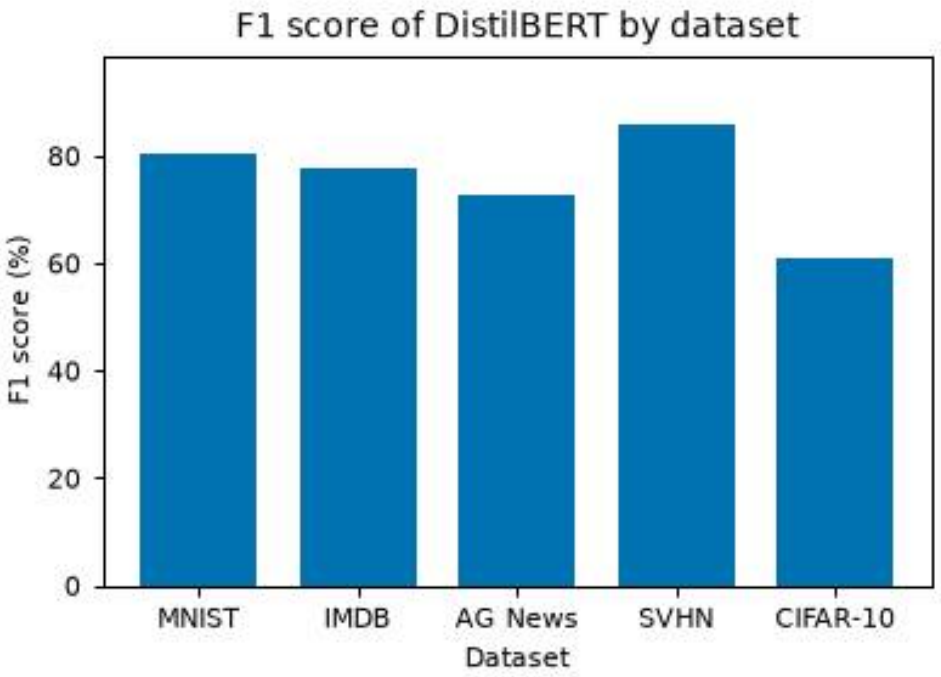


Figure 1: F1 score of DistilBERT by dataset for the models in this experiment.

DistilBERT achieves 80.5% F1 score on IMDB (Figure 1). DistilBERT scores 54.6 points higher on SVHN than on MNIST.

Figure 2 shows the f1 score by dataset.

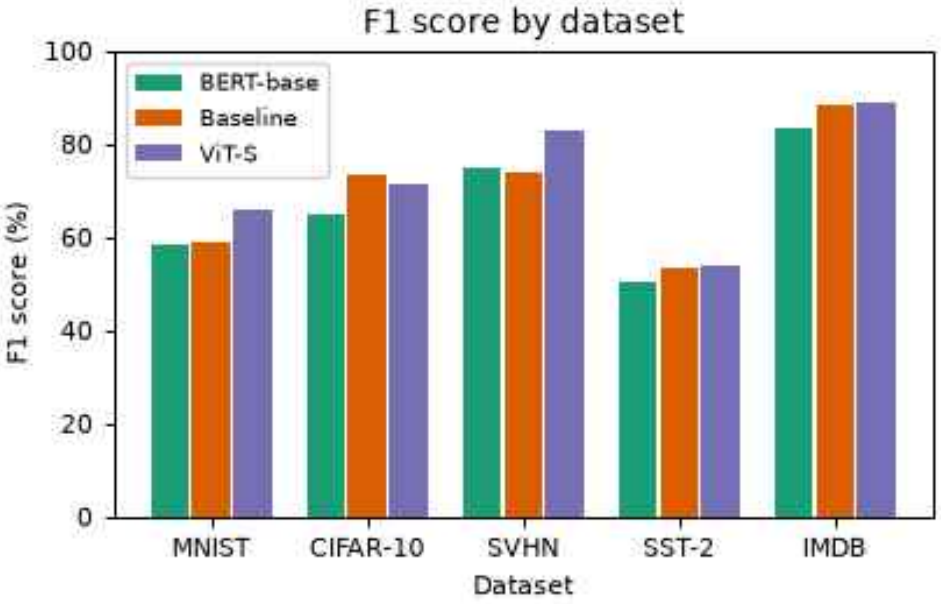


Figure 2: F1 score by dataset for the models in this experiment.

ViT-S outperforms BERT-base by about 5 percentage points on IMDB. BERT-base achieves 73.9% F1 score on SVHN (Figure 2). On MNIST, BERT-base obtains a F1 score of 65.8% (Figure 2).

Figure 3 shows the accuracy of GRU by dataset.

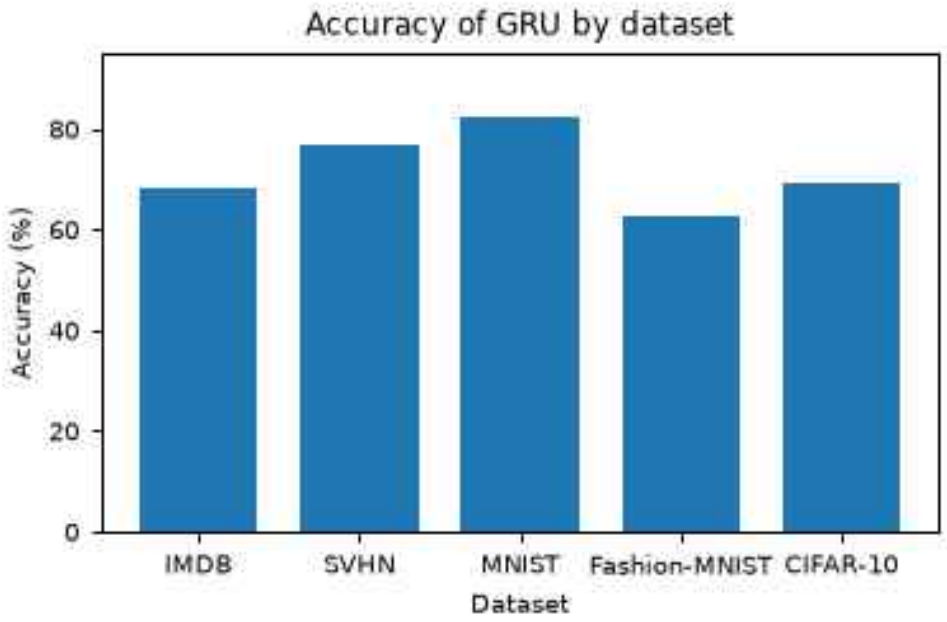


Figure 3: Accuracy of GRU by dataset for the models in this experiment.

GRU achieves 84.4% accuracy on MNIST. GRU scores 8 points higher on SVHN than on CIFAR-10. As shown in Figure 3, GRU reaches over 60% accuracy on Fashion-MNIST.

3 Discussion

On Yelp, XGBoost obtains an accuracy of 75.6%. Overall, the results are in line with what we expected from the literature.